

Case Study Assignment1

The dataset Billionaires.csv provides information on the wealth, industries, demographics, and countries of the world's billionaires. You will be using it to apply quantitative research methods to explore global patterns of wealth distribution, industry representation, and demographic differences.

Your task in this assignment is to use this dataset to formulate questions, apply statistical methods, and interpret findings that can inform discussions about technology, economics, and society. Below are the descriptions of the variables/fields in this dataset:

- **rank**: The ranking of the billionaire in terms of wealth.
- **finalWorth**: The final net worth of the billionaire in U.S. dollars.
- **category**: The category or industry in which the billionaire's business operates.
- **personName**: The full name of the billionaire.
- **age**: The age of the billionaire.
- **country**: The country in which the billionaire resides.
- **city**: The city in which the billionaire resides.
- **source**: The source of the billionaire's wealth.
- **industries**: The industries associated with the billionaire's business interests.
- **countryOfCitizenship**: The country of citizenship of the billionaire.
- **organization**: The name of the organization or company associated with the billionaire.
- **selfMade**: Indicates whether the billionaire is self-made (True/False).
- **status**: "D" represents self-made billionaires (Founders/Entrepreneurs) and "U" indicates inherited or unearned wealth.
- **gender**: The gender of the billionaire.
- **birthDate**: The birthdate of the billionaire.
- **lastName**: The last name of the billionaire.
- **firstName**: The first name of the billionaire.
- **title**: The title or honorific of the billionaire.
- **date**: The date of data collection.
- **state**: The state in which the billionaire resides.
- **residenceStateRegion**: The region or state of residence of the billionaire.
- **birthYear**: The birth year of the billionaire.
- **birthMonth**: The birth month of the billionaire.
- **birthDay**: The birth day of the billionaire.
- **cpi_country**: Consumer Price Index (CPI) for the billionaire's country.
- **cpi_change_country**: CPI change for the billionaire's country.
- **gdp_country**: Gross Domestic Product (GDP) for the billionaire's country.
- **gross_tertiary_education_enrollment**: Enrollment in tertiary education in the billionaire's country.
- **gross_primary_education_enrollment_country**: Enrollment in primary education in the billionaire's country.
- **life_expectancy_country**: Life expectancy in the billionaire's country.
- **tax_revenue_country_country**: Tax revenue in the billionaire's country.

- **total_tax_rate_country:** Total tax rate in the billionaire's country.
- **population_country:** Population of the billionaire's country.
- **latitude_country:** Latitude coordinate of the billionaire's country.
- **longitude_country:** Longitude coordinate of the billionaire's country.

Research Problem

What quantitative factors influence billionaire wealth, and how does it differ across industries, regions, and demographics?

Potential Research Questions (Choose 3-4)

1. Do billionaires in the technology industry have higher average net worth than those in other industries?
2. Is there a relationship between age and net worth among billionaires?
3. Does the number of billionaires per country correlate with GDP size?
4. Are there significant gender differences in billionaire wealth?
5.

Potential Use Cases

- **Wealth distribution analysis:** Explore the distribution of billionaires' wealth across different industries, countries, and regions.
- **Demographic analysis:** Investigate the age, gender, and birthplace demographics of billionaires.
- **Self-made vs. inherited wealth:** Analyze the proportion of self-made billionaires and those who inherited their wealth.
- **Economic indicators:** Study correlations between billionaire wealth and economic indicators such as GDP, CPI, and tax rates.
- **Geospatial analysis:** Visualize the geographical distribution of billionaires and their wealth on a map.
- **Trends over time:** Track changes in billionaire demographics and wealth over the years.

Methods

- Descriptive Statistics: Mean, median, variance, frequency distributions.
- Comparative Analysis: T-test, ANOVA, or Chi-square (for group differences).
- Correlation/Regression: Exploring relationships (e.g., age vs. net worth, GDP vs. billionaire count).
- Visualization: Histograms, bar charts, scatter plots, or boxplots to communicate results.

Tasks and Deliverables:

1. Formulate Hypotheses (e.g., *"Billionaires in technology have significantly higher net worth than those in retail"*).
2. Select 3-4 quantitative research questions from above or propose your own.
3. Analyze the data using statistical methods.
4. Present results in tables and charts.
5. Write a report (4-5 pages) with
 - a. Introduction (problem + research questions)
 - b. Methods (how you analyzed the data)
 - c. Results (tables, charts, statistics)
 - d. Discussion (interpretation + limitations of dataset)

Grading (100 points)

Criteria	Points
Clear research questions & hypotheses	15
Appropriate choice of quantitative methods	25
Correct analysis & statistical results	25
Quality of visualizations (clarity, relevance)	20
Interpretation & discussion of findings	15